Olorundara Akojede

CURRICULUM VITAE

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RESEARCH INTERESTS

Biomedical instrumentation; needs-driven medical-device design; embedded sensing and control; experimental evaluation of resource-constrained systems; edge-inference energy; global health technology.

EDUCATION

Bachelor of Science in Biomedical Engineering
University of Lagos, Lagos, Nigeria · August 2024

- Graduated as a member of the Department of Biomedical Engineering's inaugural BSc cohort.
- Undergraduate thesis: *AI Quilliri*, a team-developed healthcare language-model application incorporating privacy, security and responsible-use considerations.

RESEARCH AND ENGINEERING EXPERIENCE

Independent Researcher — Energy Measurement for Edge Inference

Lagos, Nigeria · 2026–present
Co-authors: E. V. Kwuelum and T. E. Hamzat, University of Lagos

- Designed a controlled experiment examining whether neural-network operation counts predict board-input energy across a Linux single-board computer and a microcontroller-class platform.
- Trained a 43-model corpus spanning four inference tasks and converted the selected checkpoints into 86 FP32 and full-integer INT8 TensorFlow Lite artifacts.
- Developed deployed-graph profilers for operation count, activation volume, parameter footprint, estimated tensor liveness and a static byte-access proxy; completed 10,320 fixed-bank validation invocations before energy measurement.
- Identified prespecified standard/depthwise model pairs with MAC differences of 0.203% and 1.074% but activation-output ratios of 3.975× and 3.889×, establishing controlled conditions for the hardware phase.
- Built synchronized board-input measurement workflows around an INA226 current monitor and 2 mΩ shunt: LiteRT/XNNPACK inference on a Raspberry Pi 4 and TensorFlow Lite Micro/ESP-NN inference on an ESP32-S3.
- Completed 387 accepted trials: 258 on the Raspberry Pi and 129 on the ESP32-S3. MAC count tracked broad energy scaling, but closely matched standard/depthwise pairs differed substantially in measured energy on both platforms.
- *Quantifying the Divergence Between Operation-Based Complexity Metrics and Measured Energy Consumption in Edge Inference* — accepted for presentation at MIRG ICAIR 2026.
- Technical case study · Study repository

Independent Researcher — Execution-Verifiable Safety Evaluations for Tool-Using AI Agents

Lagos, Nigeria · 2026–present

- Developing an evaluation framework that compares transcript-based safety judgments with independently recorded tool attempts, executed effects and environment-state changes.
- Specified deterministic state transitions, append-only execution records and safety predicates for authorization, information access, temporal policy compliance and report fidelity.
- Current phase: research design and evaluator specification.

Research and Development Intern — Summer Experience in Engineering Design

Oshman Engineering Design Kitchen, Rice University, Houston, Texas · May–July 2024
Team FlowMetrics · Client: Dr Mark A. Torres, Department of Earth, Environmental and Planetary Sciences

- Co-developed AquaBot during a seven-week client-led engineering programme addressing infrequent surface-water sampling in the Lake Houston watershed.
- Translated the client's monthly field workflow into requirements for unattended daily collection, 0.45 µm filtration, contamination-conscious fluid handling and indexed storage of separately retained samples.
- Integrated peristaltic pumps, a filtered reservoir, stepper-driven 32-position vial plate, needle dispenser, limit-switch homing and microcontroller control within a portable prototype.
- Contributed to CAD, iterative prototyping, electronics, 3D printing and machining while documenting the system for client delivery and the OEDK design showcase.
- Technical case study · Downloadable report · Engineering-design record

Lead Researcher and Technical Lead — IncuBed

UNILAG Design Studio / Rice360 Institute for Global Health Technologies, Lagos, Nigeria · October 2022–2024

- Co-developed a neonatal-incubator prototype after a needs-assessment visit to the neonatal unit at Lagos University Teaching Hospital identified problems with the operation and maintenance of donated incubators.
- Led the prototype's technical implementation, including CAD, acrylic fabrication, embedded programming, circuit wiring and integration of its sensing, heating and ventilation subsystems.
- Implemented temperature and humidity monitoring with DHT11 and DS18B20 sensors, relay-switched thermal actuation and a battery-powered heater/fan path in the integrated internship prototype.
- Applied locally obtainable materials and design-for-manufacture considerations to a system intended for rural primary healthcare; the laboratory build was retrospectively estimated at approximately USD 500.
- Continued the design record through company incorporation and a Nigerian patent application after the internship prototype.
- Technical case study · Downloadable report · Source-code repository

Engineering Design Lead — Automated Hand-Washing and Sanitizing System

UBORA Institute for Biomedical Engineers × United Nations Economic Commission for Africa · Remote · June 2020

- Led a distributed team during a two-week biomedical engineering boot camp focused on reducing contact during hand hygiene in the COVID-19 response.
- Developed a concept using sensor-controlled water and cleaning-fluid delivery, with stored water, battery operation and a foot-pedal mechanism supporting use during utility interruptions.
- Documented the design through the UBORA open biomedical-engineering platform; the project was approved by its assigned mentor.

PROFESSIONAL EXPERIENCE

Developer Relations — Contract

Solana Foundation, Remote · January 2025–February 2026

- Built and maintained open-source developer tooling in Rust and contributed to repositories across the Solana developer ecosystem.
- Developed `solana-accountgen`, a Rust crate for generating mock Solana accounts in program tests, including account builders, PDA construction and `solana-program-test` integration; the crate exceeded 2,000 downloads.
- Developed the open-source Solana IDL Extractor for reconstructing program interface information through bytecode and transaction analysis; the repository has more than 60 stars and several forks.
- Produced technical learning materials and maintained developer knowledge resources covering Rust, Anchor, Solana programs and testing workflows.
- Published more than 100 answers on Solana Stack Exchange, reaching more than 17,000 readers; ranked among its top five contributors for 2025 and among the top 30 contributors overall.

Data Science Instructor

GOMYCODE, Nigeria · August–October 2024

- Introduced learners with different technical backgrounds to data analysis and data-science workflows using Python.
- Guided practical exercises and adapted explanations to students' prior programming and quantitative experience.

Customer Service Agent — Contract

Jumia Nigeria, Lagos · May–June 2022

- Managed inbound and outbound customer calls, including sales-related enquiries and service resolution.

Web Development Intern

CodeLagos, Lagos · October 2018

- Participated in the fifth cohort of the technology scholarship programme through its web-development track.

SELECTED SOFTWARE AND OPEN-SOURCE PROJECTS

Argus — Experimental Smart-Contract Security Auditor

2026–present · Public architecture and capability overview

- Built a hybrid audit pipeline combining deterministic repository analysis, graph-based program analysis, specialist language-model agents, execution probes, falsification and persistent evidence records.
- Designed the workflow to separate candidate generation from verification and preserve the evidence supporting each reported security finding.

Ellisium — Privacy-Preserving Trading System

2026 · Source repository

- Developed an open-source Solana dark-pool prototype using Arcium multiparty computation for encrypted order submission, private matching and on-chain settlement.
- Implemented client-side encryption, encrypted order-book instructions and an Anchor-based settlement program; the project received third prize in Arcium's `<encrypted>` hackathon side track.

Solana IDL Extractor

2025 · Source repository

- Developed a Rust tool for recovering instructions, accounts and error information from Anchor and native Solana programs using bytecode inspection, discriminator detection, symbol recovery and optional transaction simulation.

`solana-accountgen`

2025 · Source repository · Documentation

- Published a Rust library for constructing mock accounts, program-derived addresses, SPL token accounts and reusable account maps in Solana programme tests.

SELECTED TECHNICAL OUTPUTS

Conference paper

- Akojede, O. J., Kwuelum, E. V., and Hamzat, T. E. *Quantifying the Divergence Between Operation-Based Complexity Metrics and Measured Energy Consumption in Edge Inference*. Accepted for presentation at the 6th International Conference on Artificial Intelligence Research (ICAIR), 2026.

Technical case studies

- Akojede, O. J., Kwuelum, E. V., and Hamzat, T. E. *Benchmarking Operation-Count Proxies for Edge-Inference Energy*. Technical case study, 2026.
- Akojede, O., Narayan, D., Sanyaolu, M., and Bravo, B. *AquaBot: An Automated Surface-Water Sampler for Time-Series Trace-Element Monitoring*. Technical case study, 2024/2026.

- Akojede, O., Ajayi, I., and Animasahun, R. *IncuBed: Development of a Neonatal Incubator Prototype for Rural Primary Healthcare*. Technical case study, 2022–2024/2026.

Articles

- Akojede, O. (2024). "2024 SEED Internship Reflection: Academic Learning Experience." *Medium*.
- Akojede, O. (2023). "Building on the Edge: A How-To Guide on Object Detection with Edge Impulse." *DEV Community*.
- Akojede, O. (2023). "The Magic of Attention: How Transformers Improved Generative AI." *DEV Community*.
- Akojede, O. (2023). "The Anatomy of a Programmable Logic Controller: Understanding Its Design and Architecture." *Medium*.
- Akojede, O. (2023). "Programming a PLC: Working Principle and How to Write Ladder Diagrams with GX Developer." *Medium*.

PRESENTATIONS AND DESIGN SHOWCASES

- **Summer Experience in Engineering Design Showcase**, Rice University, Houston, Texas, July 2024. Presented Team FlowMetrics' integrated AquaBot prototype and engineering-design outcome.
- **13th Annual Rice360 Global Health Technologies Design Competition**, April 2023. Team Stark participant presenting the IncuBed neonatal-incubator concept and prototype.
- **UNILAG Design Studio Internship Presentation**, Lagos, December 2022. Presented the IncuBed working prototype and design progression with Team Stark.
- **UBORA Biomedical Engineering Boot Camp against COVID-19**, June 2020. Presented a sensor-controlled automated hand-washing and sanitizing system.

INTELLECTUAL PROPERTY AND ENTREPRENEURSHIP

- **Nigerian patent application NG/P/2024/89**, *A Cost-Effective Steam Neonatal Incubator for Low-Resource Regions*. Filed February 2024; acknowledged by the Patent Registry in March 2024.
- **TinyTend Limited**, founder. Incorporated in Nigeria on 20 November 2023 as a vehicle for the continued development of IncuBed; registration number 7230687.
- **Talent Stack**, team lead, 2022. Developed an employment-platform concept for the University of Lagos Hult Prize challenge.

AWARDS AND DISTINCTIONS

- **Third Place**, Solana Confidential Balances Audit Program, for blockchain-security analysis.
- **Third Prize**, Arcium's `<encrypted>` Hackathon Side Track, 2026, for Ellisium; USD 3,500 award.
- **Top-five contributor for 2025**, Solana Stack Exchange; among the top 30 contributors overall.
- **Third Runner-Up**, Hult Prize Challenge, University of Lagos, 2022, as team lead for Talent Stack.

LEADERSHIP AND TEACHING

Founder and Machine Learning Lead

Google Developer Student Club, University of Lagos · September 2023–June 2024

- Founded the chapter's machine-learning community and developed activities for members with different levels of experience.
- Co-organized the chapter's first machine-learning hackathon in collaboration with Extended Networks W.A.; the programme received 52 applications and used a two-stage technical review process.
- Coordinated mentors, industry contributors and internal and external reviewers across the competition.

Founder and Coach — Biomedical Engineering Department Football Team

University of Lagos · Two years

- Formed the department's football team and served as its coach for two years.

Team Lead

Hult Prize Foundation, University of Lagos · May 2022

- Led the Talent Stack team to third runner-up in the local 2022 Hult Prize challenge.

PROFESSIONAL AND COMMUNITY SERVICE

- **Communications and Audio-Visual Manager**, API Conference Lagos, July 2025 — coordinated the technical conditions supporting speaker–audience interaction.
- **Guest Relations Unit Lead**, GDG Lagos DevFest, November 2023 — coordinated speaker check-in, presentation readiness and liaison with the technical team.
- **Social Media Manager**, AI Saturdays Lagos, August–December 2023 — produced cohort communications, instructor introductions and weekly learning recaps.
- **Logistics Volunteer**, Polygon DevX Tour, Lagos, July 2023 — supported registration and event operations.
- **Transportation Manager**, Open Source Community Africa Festival, June 2023 — coordinated attendee transport for the University of Lagos axis.
- **Volunteer**, GDG Lagos DevFest, November 2022 — supported registration and speaker logistics.

RESEARCH ETHICS AND PROFESSIONAL TRAINING

- **Big Data and Data Science Research Ethics**, CITI Program.
- **Artificial Intelligence and Human Subject Protections**, CITI Program.
- **Generative AI with Large Language Models**, DeepLearning.AI and Amazon Web Services via Coursera, August 2023. Credential ID: ULQWYXBHD556.
- **Getting Started with AI on Jetson Nano**, NVIDIA Deep Learning Institute, June 2023. Credential ID: 7ece759a9c3c46399713280347dc9ca7.
- **Introduction to Embedded Machine Learning**, Edge Impulse via Coursera, June 2023. Credential ID: QUWJQV2N5RMS.
- **Introduction to Machine Learning**, Kaggle, March 2023.
- **SSA Students Summit**, Google Developer Student Clubs, March 2023.
- **UBORA Biomedical Engineering Boot Camp against COVID-19**, UBORA Institute for Biomedical Engineers × UNECA, 2020.
- **Google Digital Skills Training**, Haptics Nigeria, January 2018.

TECHNICAL SKILLS

Biomedical and embedded systems: needs assessment; requirements development; sensor integration; Arduino-class microcontrollers; Raspberry Pi; ESP32-S3; I²C instrumentation; relay control; INA226 power measurement; temperature and humidity sensing; benchtop system characterization.

Engineering design and fabrication: iterative prototyping; CAD; design for manufacture; 3D printing; laser cutting; machining; acrylic fabrication; circuit wiring; electromechanical integration; fluid-handling systems.

Programming and software: Rust; Python; C/C++; TypeScript; JavaScript; Solidity; HTML/CSS; Git and GitHub; Linux; technical documentation; open-source maintenance.

Machine learning and research: TensorFlow; TensorFlow Lite/LiteRT; embedded inference; model training and quantization; deployed-graph profiling; controlled experimental design; statistical analysis planning; research synthesis; reproducible technical reporting.